WEEKLY PROGRESS REPORT — WEEK 5

Stage 5: Classical Brain MRI Tumor Segmentation Method Comparison

ECE 501 Digital Image Processing | Group 10
Project: Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing
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1. Objective

The goal of Week 5 is to compare classical image-processing approaches to the segmentation of whole tumor on BraTS2020 FLAIR MRI slices. The existing global Otsu pipeline was chosen as a basis and compared to the multi-level Otsu, adaptive/local thresholding and the K means clustering techniques with k = 2,3, and 4. Performance was measured with respect to the binary ground truth of whole-tumor segmentation (segmentation label > 0) using dice score, IoU, precision, recall, predicted area

.2. Work completed

- Used a reproducible patient/slice protocol with 20 patients selected using random seed 42.
- Randomly divided the selected cohort into a development set (DEV): 5 patients / 50 slices and a held-out test set (TEST): 15 patients / 150 slices.
- Used 6 methods on each of the selected slices, creating 1,200 method–slice records (200 slices × 6 methods). Tuned adaptive-threshold parameters on the DEV set & saved the tuning (parameter) grid.
- Produced a Dice distribution boxplot and a box plot for comparison for a representative test slice, in terms of the nature of the data.
- Computed summary statistics and Wilcoxon signed rank test comparing with global Otsu for TEST-split Dice.

3. Quantitative results

TEST split (150 slices). A higher Dice and IoU means better overlaps.

Method	Mean Dice	SD Dice	Median Dice	Mean IoU	Mean Precision	Mean Recall
Global Otsu	0.4859	0.3207	0.4181	0.3858	0.4257	0.8817
Multi-Otsu (3 classes)	0.5811	0.2369	0.6508	0.4461	0.6412	0.6840
Adaptive local	0.2838	0.1302	0.2747	0.1725	0.3058	0.3641
K-means (K=2)	0.4875	0.3194	0.4217	0.3867	0.4288	0.8784

K-means (K=3)	0.5815	0.2356	0.6492	0.4461	0.6449	0.6802
K-means (K=4)	0.5659	0.2172	0.6009	0.4245	0.7889	0.5101

The DEV selection check result showed K-means (K=3) with the highest mean Dice value (0.6069) just slightly higher than Multi-Otsu (0.6063). K-means (K=3) also had the highest TEST mean Dice (0.5815), narrowly ahead of Multi-Otsu (0.5811). The TEST split was not used to select the method.

4. Paired statistical comparison against Global Otsu

Each method alternative was compared to Global Otsu using per-slice TEST Dice values, and the results of a paired Wilcoxon signed-rank test for each were reported. These are two-sided p-values; neither of them necessarily shows which method is better so they must be read in conjunction with the values observed on each respective side.

Method vs Global Otsu	Wilcoxon statistic	p-value	Descriptive TEST mean Dice
Multi-Otsu (3 classes)	2799.5	5.36×10^{-7}	Higher (0.5811 vs 0.4859)
Adaptive local	1678.0	5.85×10^{-13}	Lower (0.2838 vs 0.4859)
K-means (K=2)	1826.5	1.98×10^{-7}	Slightly higher (0.4875 vs 0.4859)
K-means (K=3)	2838.5	7.94×10^{-7}	Higher (0.5815 vs 0.4859)
K-means (K=4)	3883.0	3.78×10^{-3}	Higher (0.5659 vs 0.4859)

Interpretation: All reported p-values are less than 0.05 for all of the alternatives, but there is variation in practical improvement. For instance, the mean-Dice increase of K-means K=2 is very small compared to that of Otsu. In the absence of a multiple-testing correction, these p-values should be interpreted as exploratory.

5. Main observations

- The best results in terms of mean Dice were the same method on both DEV and TEST, K-means K=3, with 0.0004 for the advantage of K-means over Multi-Otsu for TEST.
- The two techniques in this experiment are not so different, but rather essentially close. Multi-Otsu and K-means K=3 even surpassed the mean 'TEST' Dice produced by the global Otsu by about 0.0952 and 0.0956 respectively.
- The mean Dice and mean IoU proved that adaptive local thresholding was the least effective tested method. It is actually reported in the negative form and not as an improvement.
- The mean recall of Global Otsu and K-means K=2 was better than the mean precision, as was expected seeing that the number of pixels predicted was higher than the number of pixels in the ground-truth tumor.
- The mean precision on K-Means K=4 was more than K=5 and the mean recall was less which shows a more conservative prediction.
- High deviations in performance between slices, and zero minimum Dice indicates that there is a range of performance that has to be matched to the individual qualitative example, and puts into question the general attributes of performance.

6. Limitations and points to verify

- Calculations are done for 2D slices, not for 3D tumor volumes, and there is no assessment of inter-slice continuity.
- The benchmark includes 20 patients and 200 selected slices, and thus represents a good initial set, but not proof of clinical generalization.
- The current target is whole tumor.Separate assessment is not performed for edema, non-enhancing/necrotic core and enhancing tumor.
- K-means is an unsupervised clustering algorithm; although it is allowed within this project's classical-method scope, it is still a clustering method.
- Describe it accurately, not to say there's no algorithm like at all in the whole project. Make sure that the split and slice-selection protocol is exactly as recorded in results/selection_protocol.md, and that no TEST outcomes were used for tuning thresholds or for selecting parameters.
- The uploaded TEST type of report includes mean/standard deviation/min/max as well as paired tests. Keep the CSV per slice as the source of the reported aggregate metrics that can be audited.

7. Planned next steps

- Freeze K-means k=3 using DEV mean Dice and report its TEST performance without further tuning on TEST.
- Finish up Stage 6 error analysis: Dice vs ground truth tumor area and position of tumor range; inspect best, median and worst slices in TEST and visualize false positives and false negatives.
- Finally perform a modality comparison with the same predetermined comparison mode and protocol, but leave the decision criterion to the chasing for points unchanged.
- If time and project scope allow, go through those stages and do morphology and robustness experiments at this stage.

8. Files generated / evidence

- results/stage5_per_slice.csv — per-slice, per-method metrics.
- results/stage5_dev_metrics.csv — DEV summary statistics.
- results/stage5_test_metrics.csv — TEST summary statistics and Wilcoxon results.
- results/adaptive_tuning_grid.csv — adaptive-threshold DEV tuning grid.
- results/stage5_dice_boxplot.png — TEST Dice distribution by method.
- results/stage5_qualitative_median_slice.png — qualitative comparison on a representative TEST slice.
- results/audit_report.md — audit findings for Stages 1–4 and refactoring.

Overall conclusion

The main goal of week 5 was successfully accomplished — a repeatable comparison of six classical segmentation methods applied to a held-out multi-patient slice set. The provisional number used by DEV mean Dice is K=3 with the highest TEST mean Dice, but with a very limited advantage against Multi-Otsu. A configuration of adaptive thresholding did not improve performance. Instead of introducing more methods without justification, the next step is to analyse the errors and be transparent in the reporting of failure cases.